Zhijing Wu

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Education

Sichuan University Chengdu, China
B.Eng. in Computer Science *Sep. 2025 – Expected Jun. 2030*

- Overall GPA: **3.88/4.00**; Compulsory-course GPA: **4.00/4.00**; Compulsory-course average: **94.7/100**; Rank: **2/288**.

Research Interests

LLM Agents and Multi-Agent Systems; Agent Reliability and Evaluation; Graph-Structured Reasoning; ML Systems and Efficient Execution.

Publications

Jinghong Tang, **Zhijing Wu**, Yikun Liu, Wei Ju, and Jiyao Wang. "Hyperbolic Self-Supervised Pretraining for Hierarchical EEG Representation Learning in Generalizable Parkinson's Disease Detection." *2026 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2026).* **Regular Paper, Accepted**.

Selected Research Projects

Execution-Semantics Audit of Guided Topology Diffusion | *Independent Multi-Agent Study* 2026

- Traced generated communication graphs through diffusion supervision, runtime graph construction, message execution, and final aggregation in the released GTD pipeline on GSM8K.
- Identified a node-coordinate mismatch between generated graph representations and executed communication edges. Across **603 records**, mean edge Jaccard increased from **0.152 to 0.863** after controlled coordinate canonicalization; downstream task-performance gains were not established.

Communication-Cost and Workflow-Latency Calibration | *Independent ProtocolBench Study* 2026

- Audited ProtocolBench/AgentProtocols routing and profiled A2A, ACP, ANP, and Agora through real local RPC paths across 1–100 KiB payloads, completing **480/480 measured requests** under controlled communication workloads.
- Evaluated **120 no-op workflows (480 RPCs)** under serial and fork/join execution. Observed a within-session discrepancy between predicted and measured workflow-latency differences (predicted: −0.156 ms; observed: +0.828 ms), with execution-order-sensitive parallel results.

Trust-Metadata Inheritance in Continually Updated LLM Memory | *Independent Follow-up Study* 2026

- Audited ExpeL memory-update semantics and implemented a deterministic inherit-vs.-reset ablation of EDIT support metadata. Resetting inherited support moved an edited rule from **rank 2 to rank 15** while preserving identical rule content.
- Conducted a controlled **50-task paired WebShop evaluation**; downstream success effects remained inconclusive (**0/50 vs. 1/50**), with trajectory-observability and evaluation-budget limitations.

Within-Step Corruption Timing for Computer-Use Agents | *Independent AgentHijack Extension* 2026

- Designed controlled environment perturbations at pre-observation, post-observation/pre-execution, and post-execution boundaries, using deterministic changes and trajectory-level logging.
- Across **3 Chrome tasks and 36 primary runs**, post-observation/pre-execution perturbations increased mean episode length on all three tasks (**+4.00, +1.75, +1.00 steps**), suggesting a possible stale-action failure mode without establishing a specification violation.

Awards & Honors

First Prize (Sichuan Division), National College Student Mathematics Competition, 2026.

Meritorious Winner (M Award), Mathematical Contest in Modeling (MCM), May 2026.

Technical Skills

Programming / ML: Python, C/C++, PyTorch, PyTorch Geometric, Transformers
Graph / Agents: GCN, GraphSAGE, GAT, Message Passing, Multi-Agent Systems, Trajectory Analysis
Systems / Evaluation: Linux, Git, Docker, Conda, RPC Profiling, Controlled Experiments